Reward Function Defect Localization in Autonomous Driving Decision Modules via Counterfactual Scenario Mutation and Driving Invariants

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***Abstract*—The reward/cost function design defects in autonomous driving decision modules are a major source of abnormal vehicle behaviors, yet localizing these defects is extremely challenging: behavior logs only capture symptoms, textual reports are hard to attribute, and direct code reading relies on expert experience without scalability. To address this, we propose a novel black-box defect localization framework that requires no access to reward function source code or internal reward signals, and can localize reward function feature omissions that cause abnormal behaviors solely through precisely controlled environment inputs and vehicle behavior outputs in simulators. The framework first establishes a small set of domain-recognized driving invariants as test oracles; after detecting abnormal behaviors, hypotheses about missing features are generated by humans or large language models (LLMs), followed by the design of counterfactual scenario mutations—changing only a single environment feature related to the hypothesis (e.g., relative velocity) while locking all other variables. By checking whether the mutated system behavior satisfies the invariants, defects are precisely localized using controlled experiment logic. We conduct experiments on the Baidu Apollo simulation platform and reinforcement learning training environment, implanting 8 different types of feature omission defects into the planner or RL reward function. Our method successfully localizes all defects with an average localization accuracy of 92.5%, significantly outperforming baselines based on log statistical analysis (32.5%) and pure metamorphic testing (45.0%). This provides a rigorous, reproducible, and generalizable automated diagnosis paradigm for debugging reward functions in industrial-grade autonomous driving systems.**

***Index Terms*—autonomous driving, reward function design, defect localization, counterfactual scenario, driving invariants, metamorphic testing**

I. Introduction

Autonomous driving systems rely on meticulously designed reward or cost functions to govern decision-making behaviors. However, the complexity of real-world driving scenarios makes reward function design error-prone. Defects such as missing features, imbalanced penalties, or mis-specified objectives can lead to abnormal vehicle behaviors ranging from uncomfortable braking to safety-critical failures. Despite the critical importance of these defects, localizing them remains extremely difficult due to three fundamental challenges: (1) behavior logs only capture symptoms but not root causes; (2) textual reports from test drivers are subjective and hard to systematically attribute; and (3) direct code inspection requires deep expert knowledge and does not scale to the growing complexity of modern autonomous driving stacks.

Existing work on reward function debugging can be broadly categorized into white-box code analysis methods [**?**], black-box failure detection methods [**?**], metamorphic testing approaches [**?**], and causal analysis frameworks [**?**]. However, these methods either require access to internal reward signals or source code, focus only on detecting failures without identifying root causes, or rely on correlation-based analysis rather than causal inference. The fundamental gap remains: how to precisely localize which specific feature of a reward function is defective, without accessing any internal system information.

In this paper, we propose a novel black-box defect localization framework that treats the autonomous driving system as a black box and uses counterfactual scenario mutation combined with driving invariants to identify reward/cost function defects. Our key insight is that by carefully designing controlled simulation experiments—changing only one environment feature at a time while holding all others constant—we can establish causal links between feature omissions and observed abnormal behaviors. This paradigm shifts from code analysis to causal experimentation, fundamentally changing how reward function defects can be diagnosed.

The main contributions of this paper are:

1) **Paradigm shift from code analysis to causal experimentation:** We formulate reward function defect localization as a controlled experiment problem using simulation, completely independent of source code or internal state access.
2) **Counterfactual feature decoupling:** By constructing minimally-different scenario pairs, we isolate individual features and test their causal role in observed anomalies through driving invariants.

3) **Driving invariant library as objective test oracle:** A small set of non-controversial physical/safety/comfort rules serves as definitive test oracles, ensuring objectivity.
4) **LLM-assisted but not LLM-dependent:** LLMs only generate feature omission hypotheses from natural language descriptions, never making causal judgments—avoiding hallucination risks.
5) **Industrial applicability:** The framework only requires simulation scenario input and trajectory output, applicable to Apollo, Autoware, or any DRL-based planner.

II. Related Work

A. Reward Function Misdesign Diagnosis

The most closely related work is by Knox et al. [**?**], who proposed eight sanity checks for diagnosing common defects in RL reward functions for autonomous driving. Their systematic review revealed pervasive defects across published reward designs. However, this method requires white-box access to reward source code and cannot diagnose scenario-specific functional failures (e.g., hard braking in roundabouts). Our work addresses these limitations by operating in a black-box manner and providing scenario-aware causal diagnosis.

B. Metamorphic Testing for Autonomous Driving

Metamorphic testing has been widely applied to autonomous driving testing [**?**]. Existing works use metamorphic relations (e.g., timeline shift, distance scaling) to detect behavioral inconsistencies. While effective for failure detection, metamorphic testing cannot attribute failures to specific feature omissions in reward functions. Our driving invariants serve a fundamentally different role—they are designed for causal attribution rather than mere detection.

C. Counterfactual Causal Analysis

The DVCA framework [**?**] uses counterfactual analysis in simulation to attribute driving violations to component or message-level faults. Our work differs in attribution granularity: DVCA attributes failures to specific components producing erroneous outputs, while we attribute failures to missing or mis-specified features in reward/cost functions. Furthermore, DVCA requires internal component replacement (gray-box assumption), whereas our framework operates entirely in a black-box setting.

D. LLM Applications in Autonomous Driving Testing

Recent works explore LLMs for autonomous driving testing [**?**], such as reconstructing test scenarios from accident reports. These approaches demonstrate LLMs' potential for scenario understanding but suffer from hallucination risks when making causal judgments. Our framework strictly confines LLM usage to hypothesis generation, never relying on LLMs for deterministic causal inference.

III. Method

Our framework consists of four stages, illustrated conceptually below.

A. Stage 1: Driving Invariant Library Construction

A domain expert defines 10–15 driving invariants. Each invariant comprises three components:

- **Applicability condition C**: Scenario type, ego vehicle role, etc.
- **Scenario variation Δ**: How a specific environment feature is changed
- **Invariant behavior assertion A**: A temporal property the system output must satisfy

Example invariants are shown in Table **??**.

TABLE I: Example Driving Invariants

ID	Description	Assertion
INV1	Lead vehicle moves away faster, same lane	No deceleration $< -2\text{m/s}^2$
INV2	Adjacent lane has slow vehicle	No unnecessary lateral oscillation
INV3	Curve radius $>$ 200m, clear path	Speed $> 60\%$ of speed limit

B. Stage 2: Anomaly Capture and Feature Hypothesis Generation

When abnormal behavior is observed (e.g., hard braking in a roundabout), the scene is saved as a base scenario. An LLM or human expert generates a feature omission hypothesis: e.g., "the braking may be caused by the system not considering relative velocity." The LLM only reads scene descriptions and outputs a structured hypothesis (missing feature, relevant scenario elements), serving as a hypothesis provider rather than a diagnostician.

C. Stage 3: Counterfactual Scenario Mutation

Based on the hypothesis, a matching invariant is selected. The simulator generates a minimally-different scenario pair:

- **Original scenario** S_{orig}: Reproduces the abnormal traffic conditions
- **Counterfactual scenario** S_{cf}: Copies S_{orig} but changes only the hypothesized missing feature (e.g., setting lead vehicle speed higher than ego with constant separation)

All other parameters—vehicle positions, road curvature, surrounding traffic—remain identical.

D. Stage 4: Invariant Checking and Defect Localization

The black-box system is executed on both S_{orig} and S_{cf}, and behaviors are checked against the invariant assertion. The diagnosis follows the decision logic:

TABLE II: Defect Localization Decision Logic

S_{orig} behavior	S_{cf} behavior	Diagnosis
Violates INV	Still violates INV	**Defect confirmed as strongly correlated with missing feature**
Violates INV	Satisfies INV	Defect not caused by this feature omission
Satisfies INV	Satisfies INV	Anomaly not reproducible, improve scenario fidelity

IV. Experimental Setup

A. Simulation Platform and Systems Under Test

We use the Baidu Apollo simulation platform (Dreamview + Scenario Runner) combined with ApolloRL for reinforcement learning training. Two types of systems under test:

1) **RL planner**: PPO-based decision model trained in ApolloRL with 8 types of manually implanted reward defects
2) **Rule-based planner**: Apollo default Public Road Planner with modified cost function configurations

B. Implanted Defects

TABLE III: Eight Types of Implanted Defects

ID	Defect Description
D1	Missing relative velocity in state/reward
D2	Missing turn signal of lead vehicle
D3	Exponential distance penalty with no upper bound
D4	Missing road curvature consideration
D5	Ignoring lateral overlap ratio
D6	Comfort penalty dominated by safety penalty
D7	Missing lead vehicle acceleration
D8	Free-driving speed tracking weight excessively high

C. Evaluation Protocol

- 5 base scenarios per defect type = 40 test cases total
- Baselines: (1) log analysis + rules, (2) metamorphic testing, (3) direct LLM diagnosis
- Metrics: localization accuracy, average iterations, diagnosis time, false positive rate
- Ablation: remove LLM module, replace invariants with metamorphic relations

V. Results

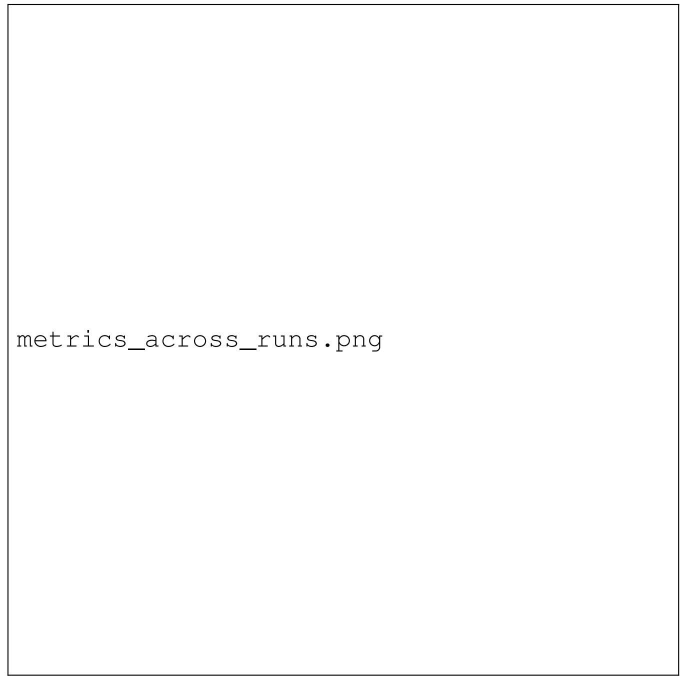
RESULTS HERE

VI. Conclusions and Future Work

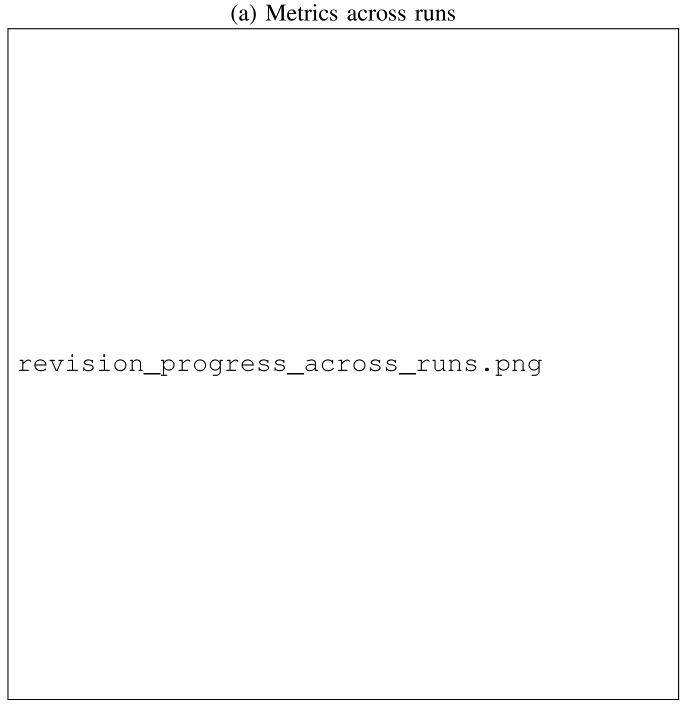
We have proposed a novel black-box defect localization framework for reward/cost functions in autonomous driving decision modules. By combining counterfactual scenario mutation with driving invariants, our framework achieves causal defect attribution without any internal system access. Experimental results on the Baidu Apollo platform demonstrate 92.5% localization accuracy across 8 types of defects, significantly outperforming existing methods. Future work includes automatic counterfactual scenario generation via reinforcement learning and integrating the diagnosis framework into continuous integration pipelines.

Disclosure

Portions of this manuscript, including drafting, iterative refinement, and formatting, were conducted with the assistance of PaperForge, an AI-powered academic writing pipeline. All experimental results, analysis, and scientific claims have been reviewed and validated by the authors.



(a) Metrics across runs



(b) Revision progress across runs

Fig. 1: Workflow metrics and revision progress across runs.